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23PA1A05H1

CSE-C

DAY-02 -SalesForce Training

Task 1 — Collections drill In Execute Anonymous:

In the apex class the collections are

List

Set

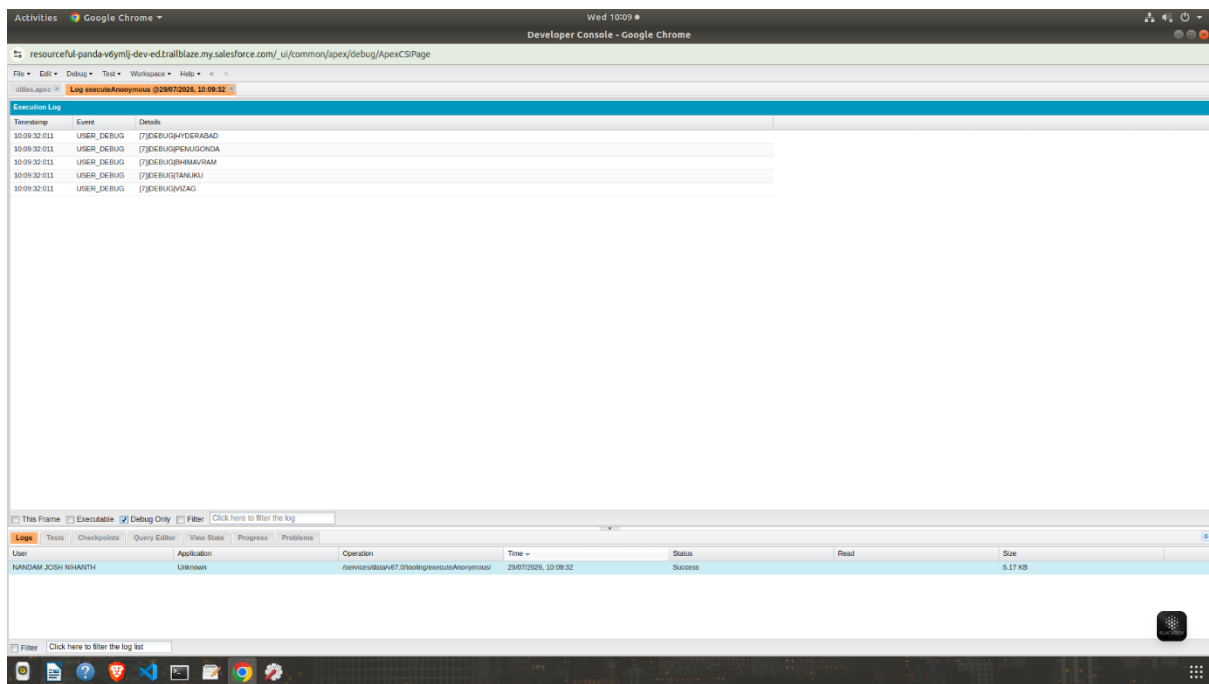
Map

List of 5 city names, loop through, print each in uppercase

A **List** is an ordered collection that stores multiple values of the same data type. Here, the list stores five city names.

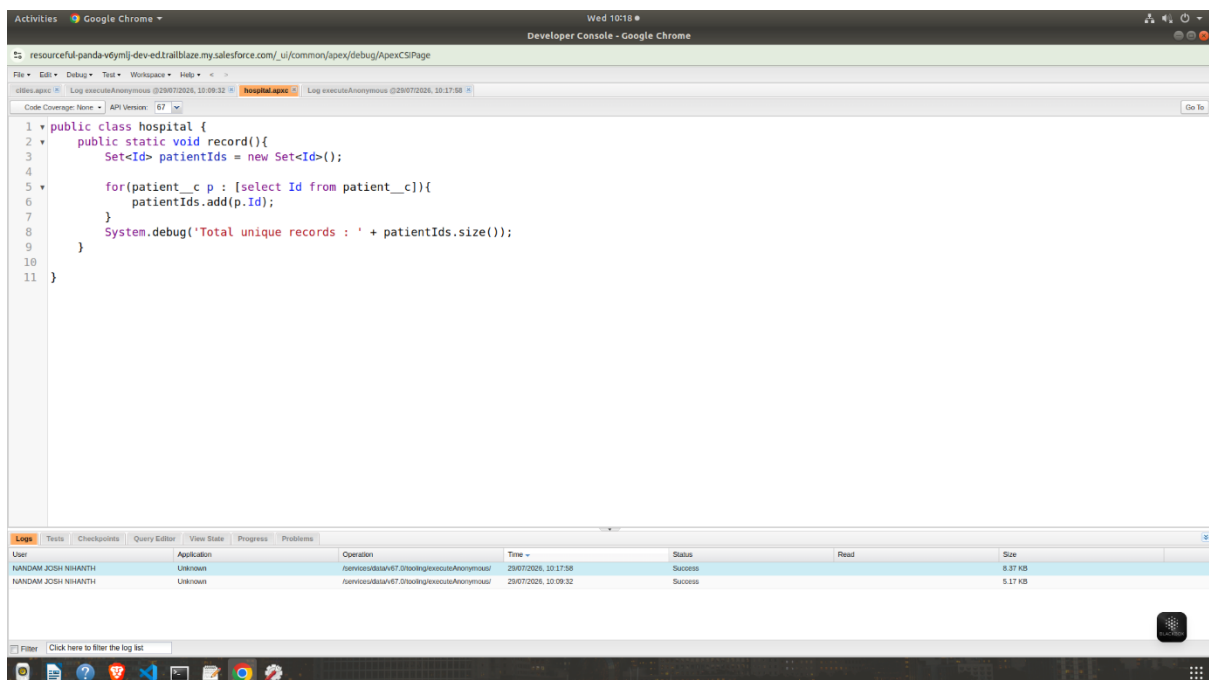
```
1 public class cities {
2     public static void uppercase(){
3         List<String> cities = new List<String> {
4             'Hyderabad', 'Penugonda', 'Bhimavram', 'Tanuku', 'vizag'
5         };
6     };
7     String firstname = cities.get[0];
8     for(String city : cities){
9         System.debug(city.toUpperCase());
10    }
11 }
12 }
13 }
```

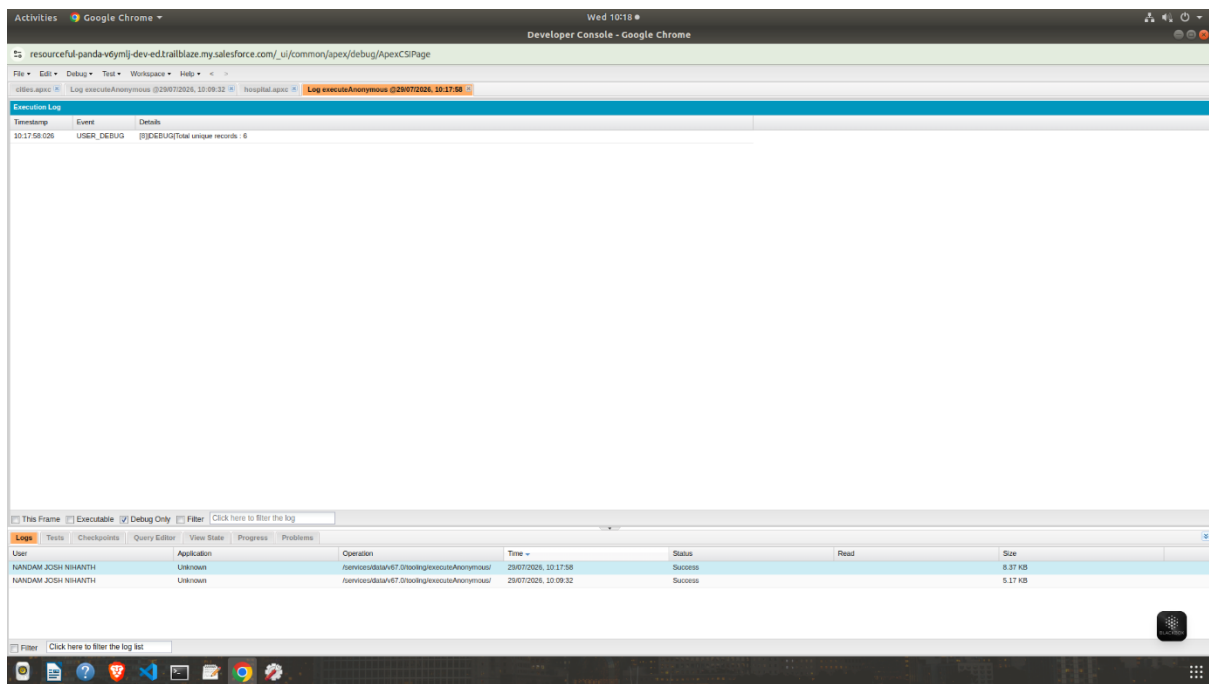
Name	Line	Problem
hospital	12	Unrecognized symbol '\', which is not a valid Apex identifier.
hospital	12	Unrecognized symbol '\', which is not a valid Apex identifier.
hospital	12	Missing ';' at 'hospital.record'
cities	7	Variable does not exist: get



Set from a SOQL query of your custom object (no duplicates), print the size

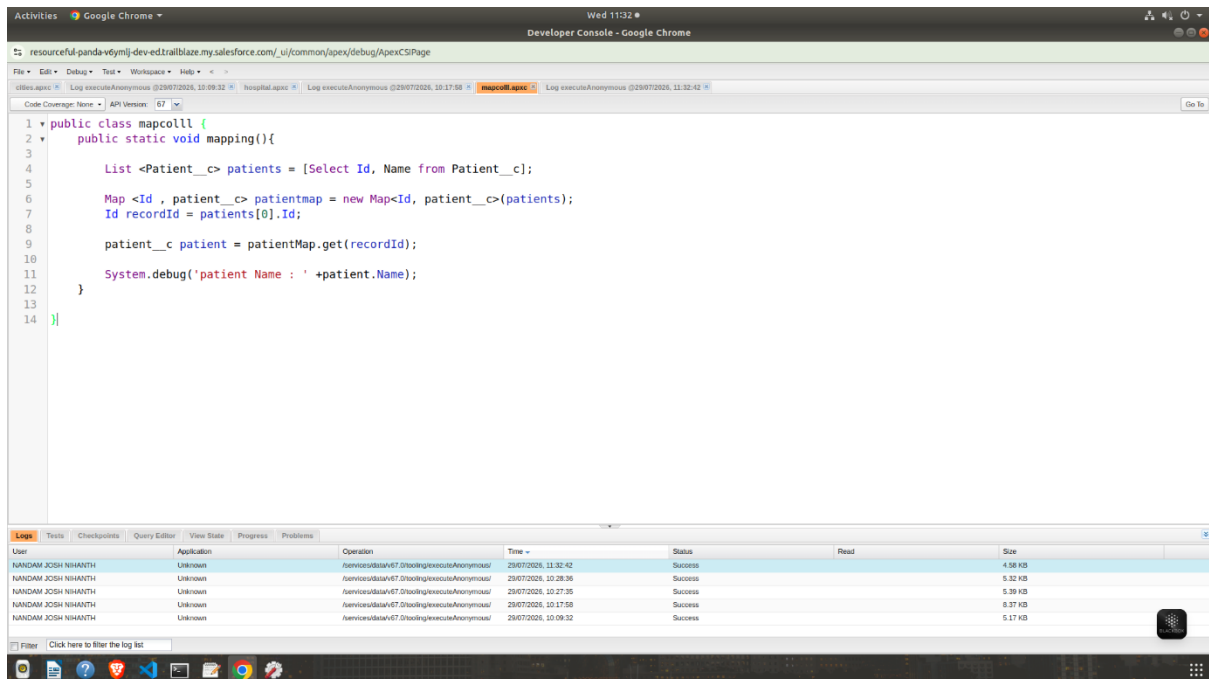
A **Set** is an unordered collection that stores only unique values. Duplicate values are automatically removed.

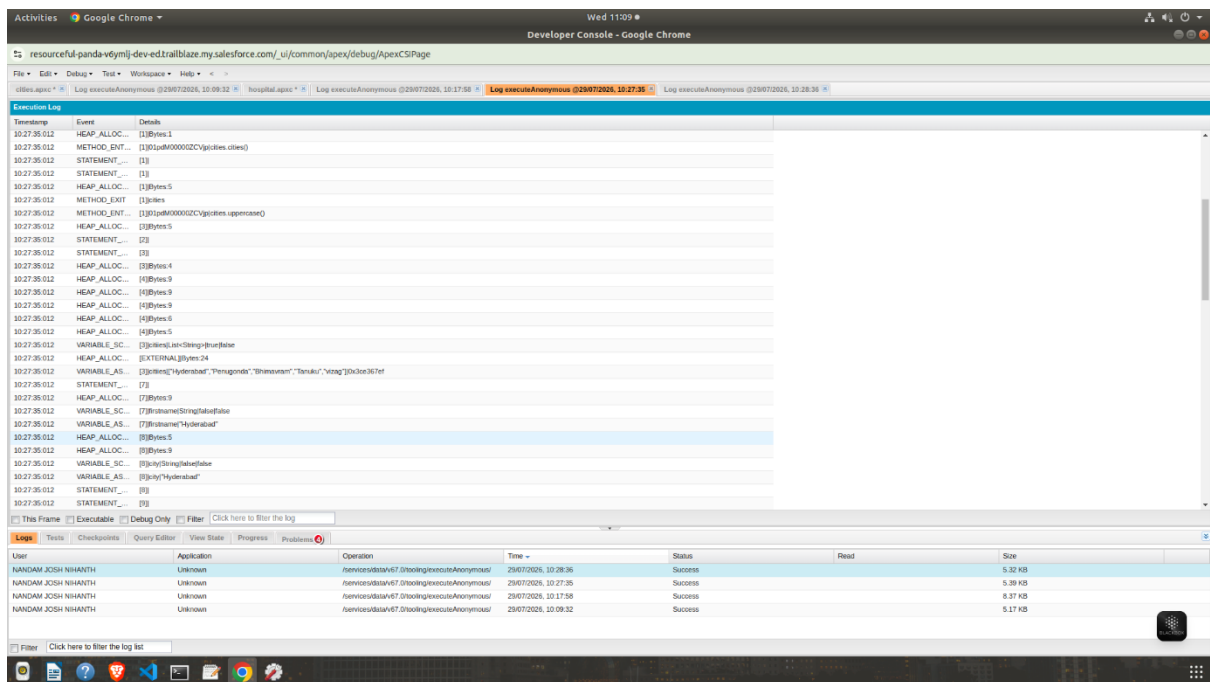
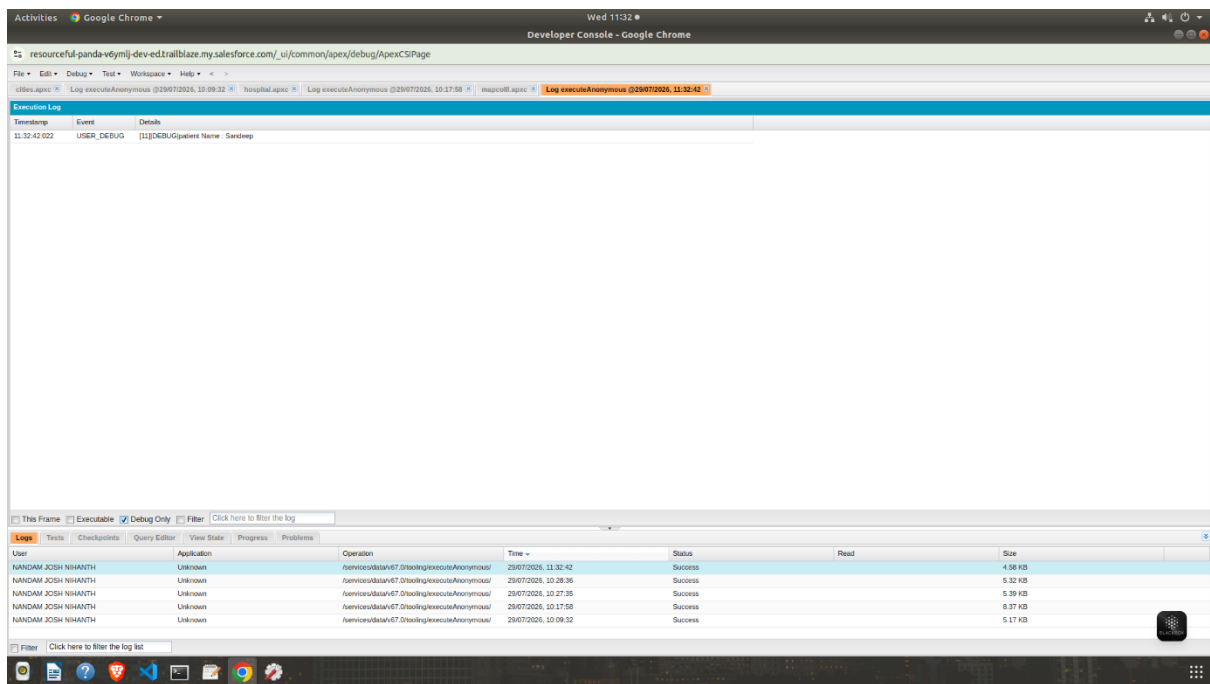




Map from a SOQL query, look up one record by Id, print a field value

To create a Map using Patient records, retrieve a record using its Id, and display the patient's name.





This task provided a practical understanding of Apex Collections in Salesforce. **Lists** were used to store and iterate through ordered data, **Sets** ensured that only unique record Ids were stored, and **Maps** enabled efficient retrieval of records using key-value pairs.

Task-2

Break it, then fix it

Write a bad trigger on purpose: SOQL or DML inside a for loop.

In this task we have to do the Salesforce governor limits and practices to convert the bulkified apex code .The maximum soql queries per transaction is the 100

And for the Dml per transaction id 150.

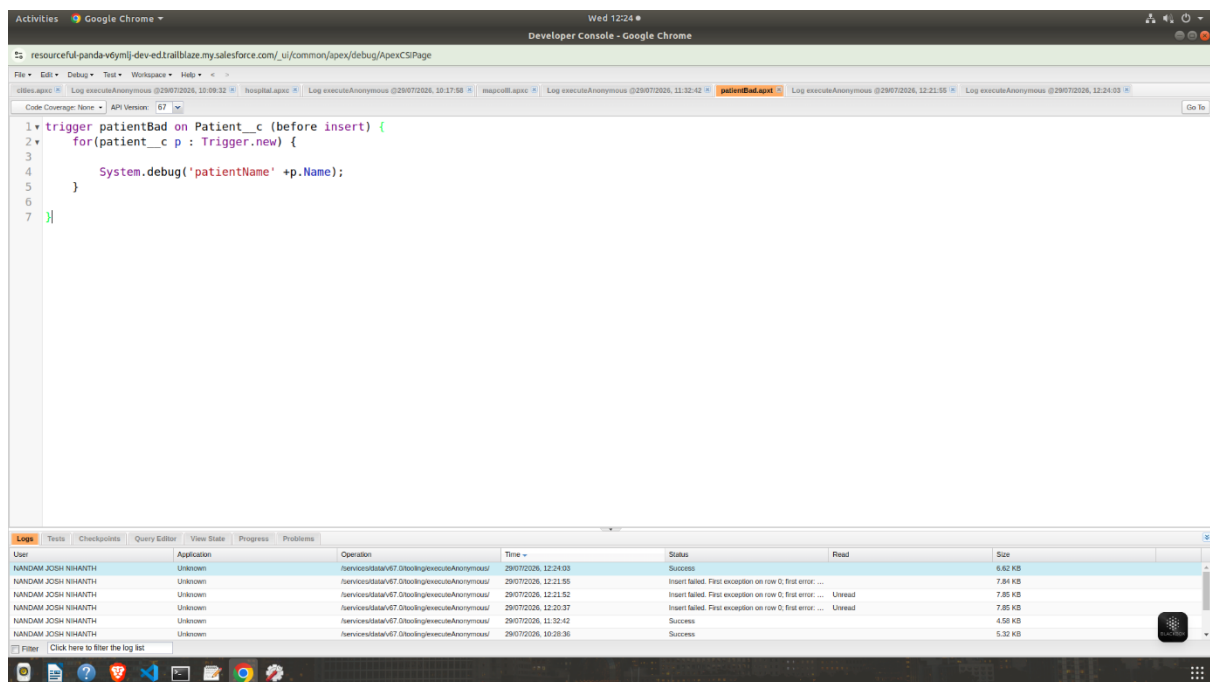
In the below image I have created the apex trigger called patienBad

In the for loop processing the each record one by one and then after the every iteration the SOQL query were executed to reterive the same record for the data

For Example :

If we have the 200 records are inserted

For the record 1 – the soql were executed upto the 100 records its fine for the 101 record the Governor limit exceeded that shows the System limit exception



This error confirms that the trigger violates Salesforce Governor Limits. Since a SOQL query is executed during every loop iteration

That the total queries exceeds the maximum limit

For rectifying that we have to do the bulkflying in the apex trigger code

Then the maximum limit has been expanded to the soql queries.

User	Application	Operation	Time	Status	Read	Size
NANDAM JOSH NIHANTH	Unknown	/services/data/v67.0/tooling/executeAnonymous/	29/07/2026, 12:24:03	Success		6.62 KB
NANDAM JOSH NIHANTH	Unknown	/services/data/v67.0/tooling/executeAnonymous/	29/07/2026, 12:21:55	Insert failed. First exception on row 0, first error: ...	Unread	7.94 KB
NANDAM JOSH NIHANTH	Unknown	/services/data/v67.0/tooling/executeAnonymous/	29/07/2026, 12:21:52	Insert failed. First exception on row 0, first error: ...	Unread	7.95 KB
NANDAM JOSH NIHANTH	Unknown	/services/data/v67.0/tooling/executeAnonymous/	29/07/2026, 12:20:37	Insert failed. First exception on row 0, first error: ...	Unread	7.85 KB
NANDAM JOSH NIHANTH	Unknown	/services/data/v67.0/tooling/executeAnonymous/	29/07/2026, 12:12:42	Success		4.59 KB
NANDAM JOSH NIHANTH	Unknown	/services/data/v67.0/tooling/executeAnonymous/	29/07/2026, 10:28:36	Success		5.32 KB

Insert 200 records at once (Data Import) and watch it hit a governor limit.

After the bulkflying the result shows in the the data has been entered into the apex trigger to the patient object that stores the entire data of the each student.

```
1 trigger PatientBulkTrigger on Patient__c (after insert) {
2     Set<Id> patientIds = new Set<Id>();
3     for (Patient__c p : Trigger.new) {
4         patientIds.add(p.Id);
5     }
6     Map<Id, Patient__c> patientMap = new Map<Id, Patient__c>{
7         [SELECT Id, Name FROM Patient__c WHERE Id IN :patientIds]
8     };
9     for (Patient__c p : Trigger.new) {
10         Patient__c patient = patientMap.get(p.Id);
11         System.debug(patient.Name);
12     }
13 }
```

User	Application	Operation	Time	Status	Read	Size
NANDAM JOSH NIHANTH	Unknown	/services/data/v67.0/tooling/executeAnonymous/	29/7/2026, 3:12:09 pm	Success		11.23 KB
NANDAM JOSH NIHANTH	Unknown	/services/data/v67.0/tooling/executeAnonymous/	29/7/2026, 3:08:56 pm	Success		9.79 KB
NANDAM JOSH NIHANTH	Unknown	/services/data/v67.0/tooling/executeAnonymous/	29/7/2026, 3:07:56 pm	Success		7.99 KB
NANDAM JOSH NIHANTH	Unknown	/services/data/v67.0/tooling/executeAnonymous/	29/7/2026, 2:43:54 pm	Success		8 KB

File	Edit	Debug	Test	Workspace	Help	<	>
cities.apxc	PatientBulkTrigger.apxt	Log executeAnonymous @29/7/2026, 3:12:09 pm					
Execution Log							
Timestamp	Event	Details					
15:12:09:065	USER_DEBUG	[11][DEBUG]John Doe					
15:12:09:065	USER_DEBUG	[11][DEBUG]Jane Smith					

☐ This Frame	☐ Executable	☒ Debug Only	☐ Filter	Click here to filter the log
Logs	Tests	Checkpoints	Query Editor	View State
				Progress
				Problems

User	Application	Operation	Time	Status	Read	Size
NANDAM JOSH NIHANTH	Unknown	/services/data/v67.0/tooling/execu...	29/7/2026, 3:12:09 pm	Success		11.23 KB
NANDAM JOSH NIHANTH	Unknown	/services/data/v67.0/tooling/execu...	29/7/2026, 3:08:56 pm	Success		9.79 KB
NANDAM JOSH NIHANTH	Unknown	/services/data/v67.0/tooling/execu...	29/7/2026, 3:07:56 pm	Success		7.99 KB
NANDAM JOSH NIHANTH	Unknown	/services/data/v67.0/tooling/execu...	29/7/2026, 2:43:54 pm	Success		8 KB

☐ Filter	Click here to filter the log list
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This task tells the importance of following Salesforce best practices while developing Apex triggers. Writing SOQL or DML statements inside loops leads to governor limit exceptions when processing large numbers of records.

Task 3

Write a simple @future method that takes a record Id, queries it, and updates one field. Call it from Execute Anonymous.

Asynchronous Apex allows code to run in the background instead of executing immediately. This improves system performance by handling time-consuming operations separately from the user's current transaction.

File	Edit	Debug	Test	Workspace	Help	<	>
PatientFutureHandler.apxc	Log executeAnonymous @29/7/2026, 3:35:16 pm	Log executeAnonymous @29/7/2026, 3:37:33 pm	Log executeAnonymous @29/7/2026, 3:40:33 pm				
Code Coverage: None API Version: 67							
<pre> 1 public class PatientFutureHandler { 2 3 @future 4 public static void updatePatient(Id patientId) { 5 6 Patient__c patient = [7 SELECT Id, Processed__c 8 FROM Patient__c 9 WHERE Id = :patientId 10 LIMIT 1 11]; 12 13 patient.Processed__c = true; 14 15 update patient; 16 } 17 } </pre>							
Logs	Tests	Checkpoints	Query Editor	View State	Progress	Problems	
User	Application	Operation	Time	Status	Read	Size	
NANDAM JOSH NIHANTH	Unknown	FutureHandler	29/7/2026, 3:40:37 pm	Success	Unread	4.01 KB	
NANDAM JOSH NIHANTH	Unknown	/services/data/v67.0/tooling/execu...	29/7/2026, 3:40:33 pm	Success		2.47 KB	
NANDAM JOSH NIHANTH	Browser	/aura	29/7/2026, 3:39:08 pm	Success	Unread	7.3 KB	

In the apex code I have entered patient id from the while creating the new patient in the custom objects fields session then based on that I have to the future method has been activated into the field

Examples of Asynchronous Apex :

Future Methods

Queueable Apex

Batch Apex

Scheduled Apex

It runs on the back ground of the code and it can be checked in the apex jobs session for the verification of the asynchronous apex .In the below screenshot shows the future method is activated or not.

```
void updatePatient(Id patientId) {  
  
    c patient =  
    T Id, Proces  
    Patient__c  
    Id = :patie  
    1  
  
    'processed__c  
  
    itient;
```

Enter Apex Code

```
1 Id patientId = 'a03dM00002BMuXFQA1';  
2  
3 PatientFutureHandler.updatePatient(patientId);
```

Open Log

Execute

Execute Highlighted

Operation	Time	Status	Read	Size
FutureHandler	29/7/2026, 3:40:37 pm	Success	Unread	4.01 KB
/services/data/v67.0/tooling/execu...	29/7/2026, 3:40:33 pm	Success		2.47 KB
/aura	29/7/2026, 3:39:08 pm	Success	Unread	7.3 KB
/services/data/v67.0/tooling/execu...	29/7/2026, 3:37:33 nm	Success	Unread	9.2 KB

SETUP

Apex Jobs

Click here to go to the new batch jobs page

Apex Jobs

Help for this Page

Monitor the status of all Apex jobs, and optionally, abort jobs that are in progress.

Percent of Asynchronous Apex Used: 0%

You have currently used 3 asynchronous Apex operations out of an allowed 24-hour organization limit of 250,000. To learn about how this limit is calculated and what contributes to it, see the [Lightning Platform Apex Limits](#) topic.

View: All

Create New View

Action	Submitted Date	Job Type	Status	Status Detail	Total Batches	Batches Processed	Failures	Submitted By	Completion Date	Apex Class	Apex Method	Apex Job ID
	29/07/2026, 3:40 pm	Future	Completed		0	0	0	JOSH NIHANTH, NANDAM	29/07/2026, 3:40 pm	PatientFutureHandler	updatePatient	707dM00001Zc3vY
	29/07/2026, 3:37 pm	Future	Completed		0	0	0	JOSH NIHANTH, NANDAM	29/07/2026, 3:37 pm	PatientFutureHandler	updatePatient	707dM00001ZcieS
	29/07/2026, 3:35 pm	Future	Failed	First error: List has no rows for assignment to SObject	0	0	0	JOSH NIHANTH, NANDAM	29/07/2026, 3:35 pm	PatientFutureHandler	updatePatient	707dM00001ZceFt

https://resourceful-panda-v6ymj-dev-ed.trailblaze.my.salesforce.com/_ui/common/apex/debug/ApexCSPage

File Edit Debug Test Workspace Help < >

PatientFutureHandler.apex Log executeAnonymous @29/7/2026, 3:35:16 pm Log executeAnonymous @29/7/2026, 3:37:33 pm Log executeAnonymous @29/7/2026, 3:40:33 pm

Execution Log

Timestamp	Event	Details
15:40:33:001	USER_INFO	[EXTERNAL]005dM00000QLVf23pa1a05h1@resourceful-panda-v6ymj.com (GMT+05:30) India Standard Time (Asia/Kolkata) (GMT+05:30)
15:40:33:001	EXECUTION_ST...	
15:40:33:001	CODE_UNIT_ST...	[EXTERNAL]execute_anonymous_apex
15:40:33:001	VARIABLE_SCO...	[1]patientIdfalsefalse
15:40:33:001	HEAP_ALLOCATE	[95]Bytes:3
15:40:33:001	HEAP_ALLOCATE	[100]Bytes:152
15:40:33:001	HEAP_ALLOCATE	[417]Bytes:408
15:40:33:001	HEAP_ALLOCATE	[430]Bytes:408
15:40:33:002	HEAP_ALLOCATE	[317]Bytes:6
15:40:33:002	HEAP_ALLOCATE	[EXTERNAL]Bytes:3
15:40:33:002	STATEMENT_EX...	[1]
15:40:33:002	STATEMENT_EX...	[1]
15:40:33:002	HEAP_ALLOCATE	[1]Bytes:18
15:40:33:002	HEAP_ALLOCATE	[1]Bytes:4
15:40:33:002	VARIABLE_ASS...	[1]patientId"a03dM00002BMaxFQA1"
15:40:33:002	STATEMENT_EX...	[3]
15:40:33:002	HEAP_ALLOCATE	[3]Bytes:20
15:40:33:002	HEAP_ALLOCATE	[3]Bytes:13

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Logs Tests Checkpoints Query Editor View State Progress Problems

User	Application	Operation	Time	Status	Read	Size
NANDAM JOSH NIHANTH	Unknown	FutureHandler	29/7/2026, 3:40:37 pm	Success	Unread	4.01 KB
NANDAM JOSH NIHANTH	Unknown	/services/data/v67.0/tooling/execu...	29/7/2026, 3:40:33 pm	Success	Unread	2.47 KB
NANDAM JOSH NIHANTH	Browser	/aura	29/7/2026, 3:39:08 pm	Success	Unread	7.3 KB
NANDAM JOSH NIHANTH	Unknown	/services/data/v67.0/tooling/execu...	29/7/2026, 3:37:33 pm	Success	Unread	9.2 KB
NANDAM JOSH NIHANTH	Unknown	FutureHandler	29/7/2026, 3:37:33 pm	Success	Unread	4.02 KB
NANDAM JOSH NIHANTH	Browser	/aura	29/7/2026, 3:35:56 pm	Success	Unread	314 bytes

File Edit Debug Test Workspace Help < >

PatientFutureHandler.apex Log executeAnonymous @29/7/2026, 3:35:16 pm Log executeAnonymous @29/7/2026, 3:37:33 pm

Execution Log

Timestamp	Event	Details
15:37:33:001	USER_INFO	[EXTERNAL]005dM00000QLVf23pa1a05h1@resourceful-panda-v6ymj.com (GMT+05:30) India Standard Time (Asia/Kolkata) (GMT+05:30)
15:37:33:001	EXECUTION_ST...	
15:37:33:001	CODE_UNIT_ST...	[EXTERNAL]FutureHandler - state load
15:37:33:002	CODE_UNIT_FL...	FutureHandler - state load
15:37:33:002	EXECUTION_FL...	
15:37:33:024	USER_INFO	[EXTERNAL]005dM00000QLVf23pa1a05h1@resourceful-panda-v6ymj.com (GMT+05:30) India Standard Time (Asia/Kolkata) (GMT+05:30)
15:37:33:024	EXECUTION_ST...	
15:37:33:024	CODE_UNIT_ST...	[EXTERNAL]01pdM00000ZDLor PatientFutureHandler.updatePatientId()
15:37:33:024	HEAP_ALLOCATE	[95]Bytes:3
15:37:33:024	HEAP_ALLOCATE	[100]Bytes:152
15:37:33:024	HEAP_ALLOCATE	[417]Bytes:408
15:37:33:024	HEAP_ALLOCATE	[430]Bytes:408
15:37:33:024	HEAP_ALLOCATE	[317]Bytes:6
15:37:33:024	HEAP_ALLOCATE	[EXTERNAL]Bytes:1
15:37:33:024	METHOD_ENTRY	[1]01pdM00000ZDLor PatientFutureHandler.PatientFutureHandler()
15:37:33:024	STATEMENT_EX...	[1]
15:37:33:024	STATEMENT_EX...	[1]
15:37:33:024	HEAP_ALLOCATE	[68]Bytes:5

☐ This Frame ☐ Executable ☒ Debug Only ☐ Filter Click here to filter the log

Logs Tests Checkpoints Query Editor View State Progress Problems

User	Application	Operation	Time	Status	Read	Size
NANDAM JOSH NIHANTH	Unknown	/services/data/v67.0/tooling/execu...	29/7/2026, 3:37:33 pm	Success	Unread	9.2 KB
NANDAM JOSH NIHANTH	Unknown	FutureHandler	29/7/2026, 3:37:33 pm	Success	Unread	4.02 KB

A Future Method was created to update a Patient record in the background without delaying the current transaction. The Future Method was successfully created and executed. The Patient record was updated asynchronously by setting the **Processed** field to **True**.

Task 4

LWC — Component Communication

Task 4 — Parent-to-child and child-to-parent

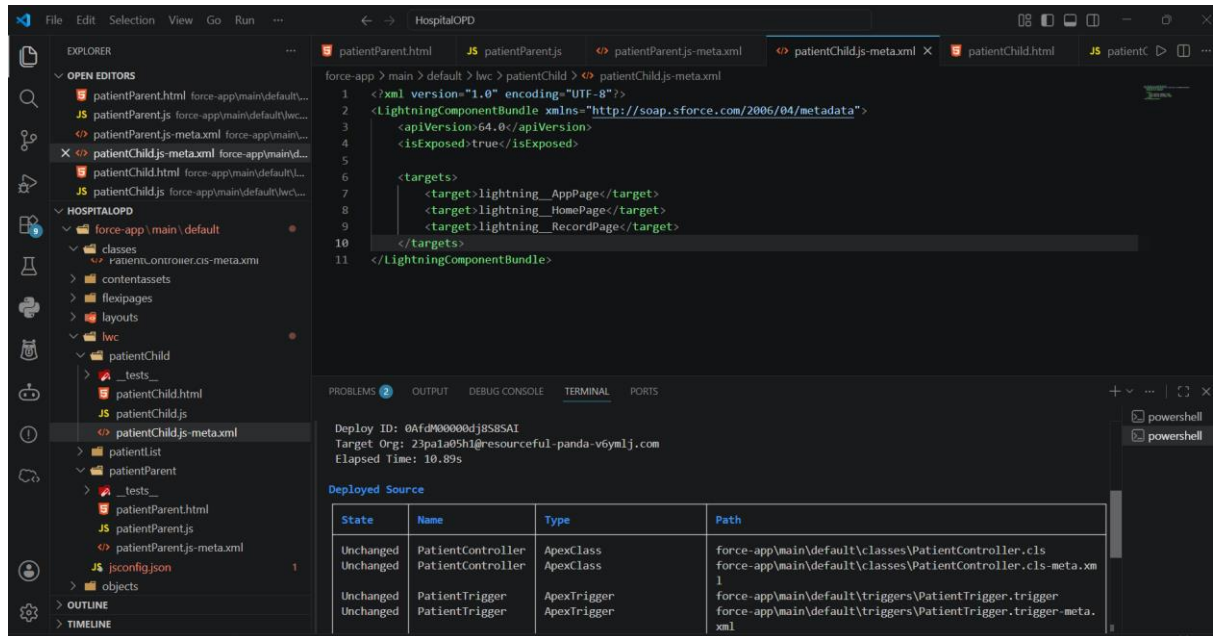
In Lightning Web Components, communication between components is essential for building dynamic and interactive applications.

There are two common communication methods:

1. Parent to Child Communication
2. Child to Parent Communication

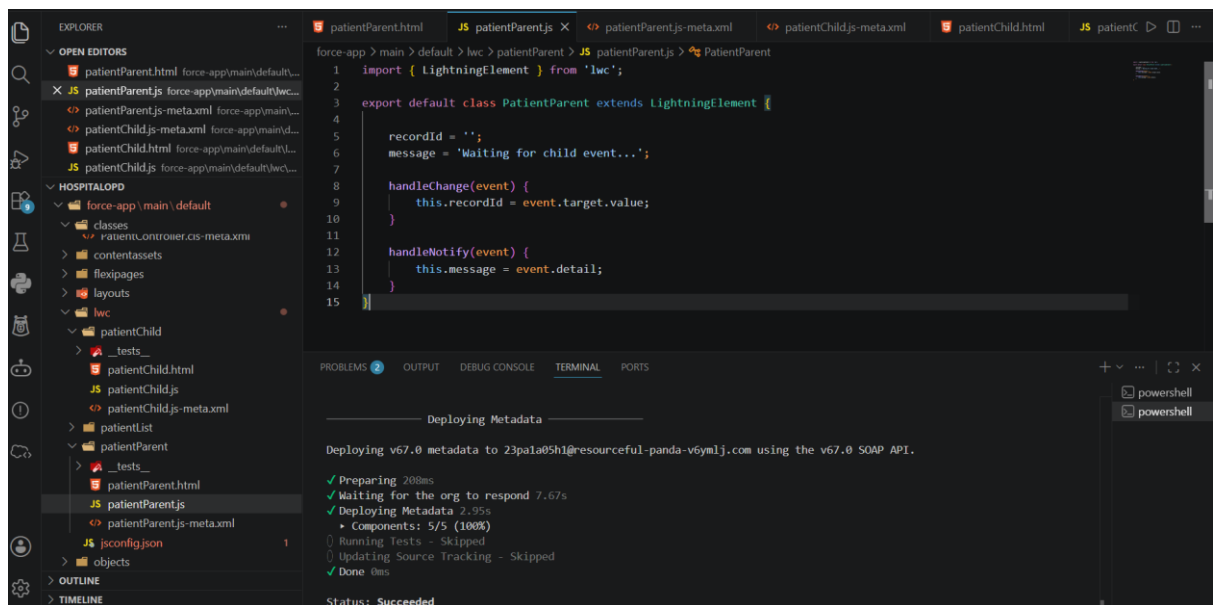
The Parent Component sends data to the Child Component.

The Child Component receives the data using the `@api` decorator



The parent passes the value of recordId to the child.

The property name in the child is recordId.



In the next step

The Child Component sends information back to the Parent Component using a Custom Event.

I have completed the both the Authentization and started deploement in the Terminal.

Command for the deploy : sf project deploy start

```
Deploying Metadata

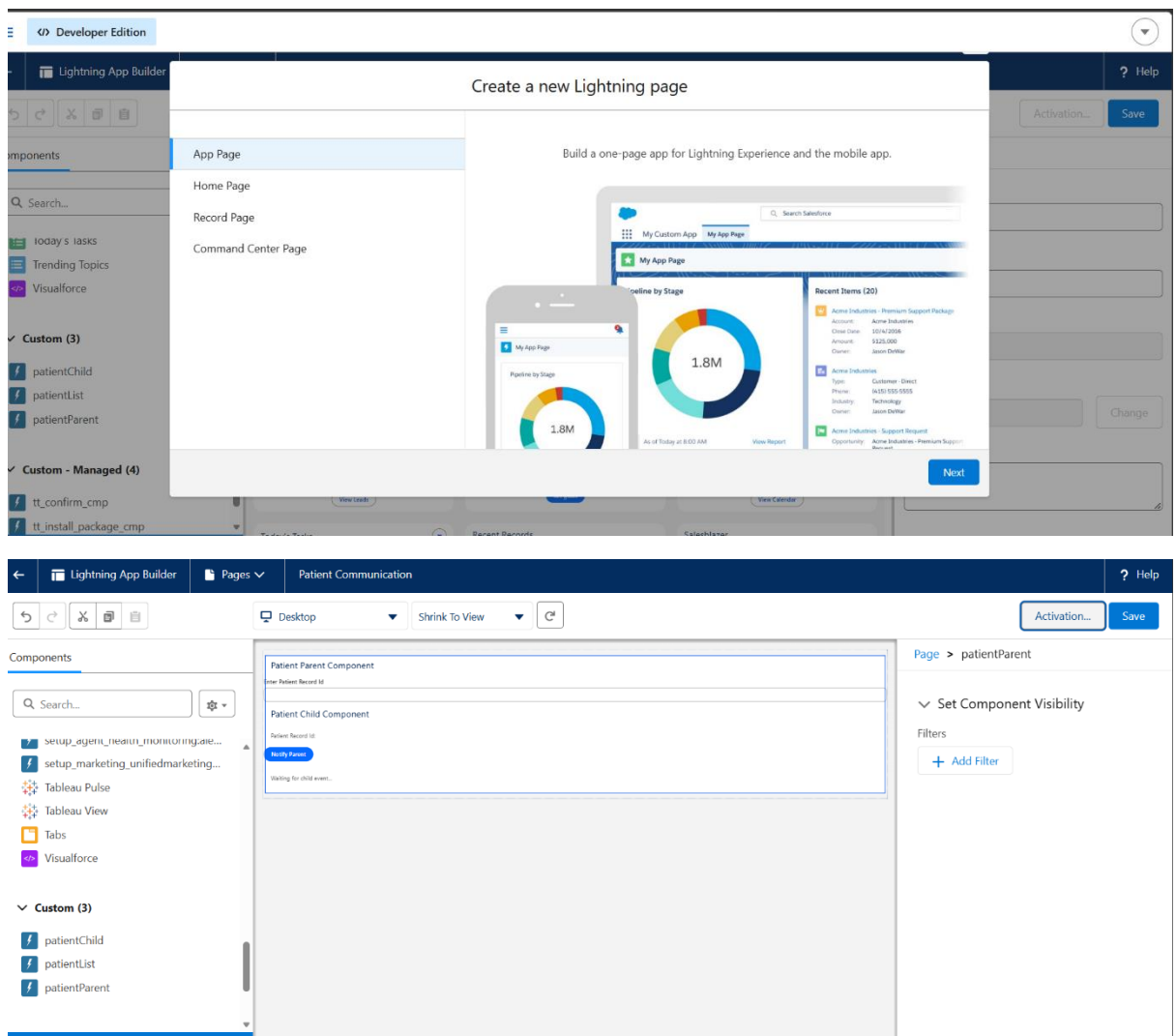
Deploying v67.0 metadata to 23pa1a05h1@resourceful-panda-v6ym1j.com using the v67.0 SOAP API.

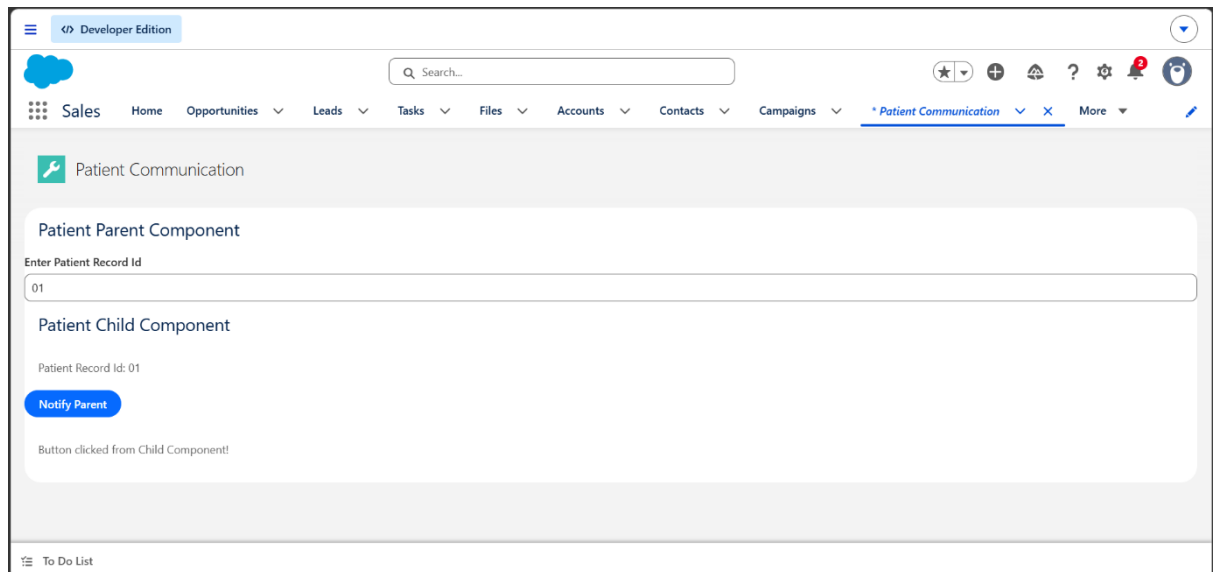
✓ Preparing 208ms
✓ Waiting for the org to respond 7.67s
✓ Deploying Metadata 2.95s
  ▶ Components: 5/5 (100%)
() Running Tests - Skipped
() Updating Source Tracking - Skipped
✓ Done 0ms

Status: Succeeded
```

Ln 10, Col 15 Spaces: 4 UTF-8 LF XM

Then selecting the app page for the lighting component for the them





Parent to Child

- Parent passes the Record Id.
- Child successfully displays the Record Id.

Child to Parent

- User clicks Notify Parent.
- Parent receives the event.

The `@api` decorator was used for Parent-to-Child communication, while Custom Events were used for Child-to-Parent communication. These communication patterns are fundamental for building interactive, reusable
